

Trust yourself. You know more than you think you do. Each question is worth 2 points except where noted.

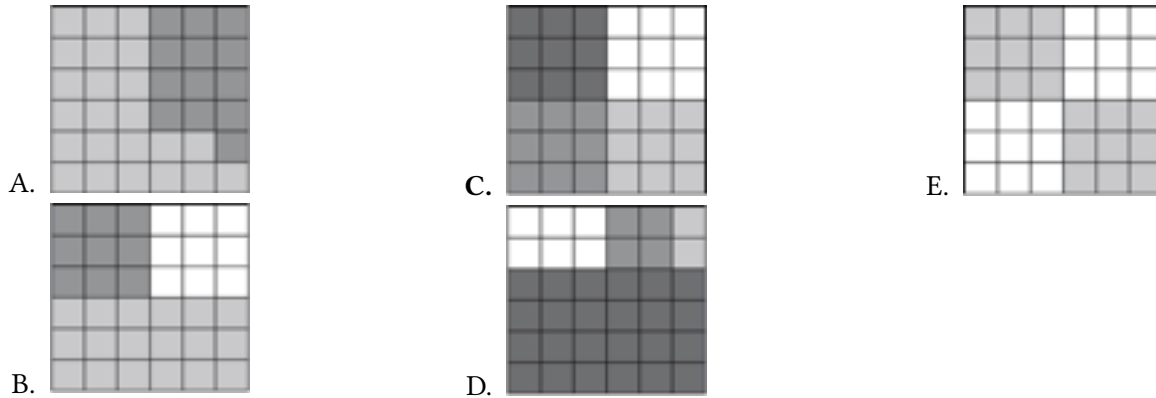
Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

1. Given the following values and assuming exponential population growth, what will be the total population size after one generation has elapsed? Round to nearest whole individual.

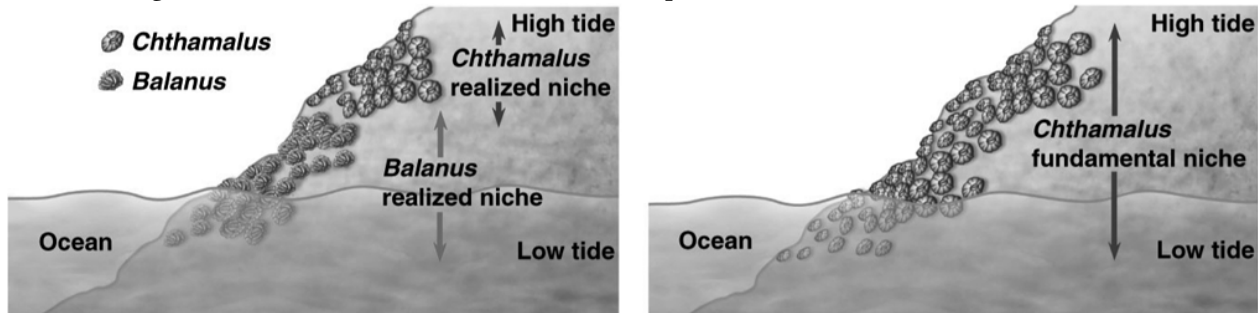
$$\begin{array}{ll} b = 0.09 & N = 18,657 \\ d = 0.12 & K = 15,000 \end{array}$$

- A. 560
 - B. 3,657
 - C. 15,000
 - D. 18,097**
 - E. 19,217
2. The attribute that most influences population density and rates of immigration and emigration among habitat patches in a metapopulation are
 - A. habitat quality, size of sink patches and isolation of source patches.
 - B. patch size, patch isolation and species diversity.
 - C. patch size, habitat quality, and isolation of sink patches.
 - D. patch isolation, habitat quality, and size of source patches.
 - E. patch size, patch isolation and habitat quality.**
 3. Which of the following is most likely to contribute to density-dependent regulation of populations?
 - A. Fires.
 - B. Intraspecific competition for nutrients.**
 - C. Floods.
 - D. The removal of toxic waste by decomposers.
 - E. Earthquakes.
 4. Which of the following statements is consistent with the principle of competitive exclusion?
 - A. The density of one competing species will have a positive impact on the population growth of the other competing species.
 - B. Bird species generally do not compete for nesting sites.
 - C. Two species with the same fundamental niche will exclude other competing species.
 - D. Evolution tends to increase competition between related species.
 - E. Even a slight reproductive advantage will eventually lead to the elimination of the less well adapted of two competing species.**
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5. Which block in the figure below is most consistent intermediate levels of disturbance in an ecosystem? (Different shades, including white, represent different species. The small squares indicate relative abundance of individuals present.)



6. In the experiment shown in the figure below, *Balanus* was removed from the habitat shown on the left. Which of the following statements is a valid conclusion of this experiment?



- A. The removal of *Balanus* shows that the realized niche of *Chthamalus* is smaller than its fundamental niche.
- B. If *Chthamalus* were removed, *Balanus*'s fundamental niche would become larger.
- C. These two species of barnacle do not show competitive exclusion.
- D. *Balanus* can survive only in the lower intertidal zone because it is unable to resist dehydration.
- E. *Balanus* is inferior to *Chthamalus* in competing for space on rocks lower in the intertidal zone.
7. Most insects have two pairs of wings. Flies only have one pair of wings. The second pair of wings is reduced to tiny structures called halteres. The mutation(s) that led to the change from wings to halteres could have
- A. altered the expression of a *Hox* gene.
- B. duplicated all of the *Hox* genes in the fly genome.
- C. caused the loss of *Antp* or *Ubx*.
- D. resulted in paedomorphosis.
- E. caused expression of a pseudogene.

8. A harmless fly looks like a dangerous wasp. The fly is using
- A. disruptive coloration.
 - B. cryptic coloration.
 - C. warning coloration.
 - D. camouflage.
 - E. mimicry.**
9. The dispersion pattern for individuals across the entire metapopulation shown in question 38 is most likely
- A. uniform.
 - B. even.
 - C. clumped.**
 - D. random.
 - E. gene flow.
10. Which of the following is *not* directly concerned with community ecology?
- A. intraspecific competition**
 - B. predation
 - C. herbivory
 - D. resource partitioning
 - E. symbiosis
11. Naturally occurring fires cause occasional disturbance to an ecosystem. If humans stop these fires as soon as they are started, then
- A. species diversity in the ecosystem will tend to decrease.**
 - B. species diversity in the ecosystem will tend to increase.
 - C. species diversity will tend to remain unchanged, but the types of species present may change.
 - D. the diversity of threatened and endangered species will be protected.
12. White-breasted nuthatches and brown creepers both eat insects that hide in the furrows of bark in hardwood trees. The brown creeper searches for insects by hunting from the bottom of the tree trunk toward the top, whereas the white-breasted nuthatch searches from the top of the trunk down. Their feeding areas rarely overlap on the tree trunk. These hunting behaviors best illustrate which of the following ecological concepts?
- A. keystone species
 - B. intraspecific competition
 - C. interspecific competition
 - D. resource partitioning**
 - E. competitive exclusion
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Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Aposomatic	Gene duplication	Mutualism
Commensalism	Habitat fragmentation	Niche compression
Corridor	Immigration	Parasitism
Cryptic coloration	Intermediate succession	Paedomorphy
Diversity	Interspecific competition	Pseudogene
Early succession	Intraspecific competition	Realized niche
Emigration	Keystone species	Resource Partitioning
Evenness	Late succession	Richness
Fundamental niche	Mimicry	Symbiosis

13. **Realized niche** Based on the resources actually used by a species.
14. **Keystone Species** A species that greatly increases community diversity.
15. **Emigration** Individuals leave a population.
16. **Aposomatic Coloration** A poisonous species warns a potential predator.
17. **Paedomorphy** Retention of child-like traits in adult organism.
18. **Intraspecific Competition** Occurs when individuals of the same species require the same limited resource.
19. **Early succession** Composed mostly of grasses and other generalist species.
20. **Pseudogene** Duplicated gene loses its function.
21. **Richness** The total number of species present in the community.
22. **Commensalism** One species gets ecological benefit over the long term, the other neither benefits or loses.
23. **Corridor** Can reduce local extinction by connecting sources and sinks.
24. (2 points E.C.) Write the name of the band or musical artist that you think is Dr. Taylor's favorite.

Circle True or False. Read each statement completely before answering.

25. **TRUE** False The carrying capacity of a habitat may change as environmental conditions change.
26. True **FALSE** Random dispersion occurs when all individuals in a population are evenly spaced.
27. True **FALSE** Evenness measures the distribution of relative abundance across multiple species in a population.
28. **TRUE** False The intrinsic rate of population growth does not include immigration or emigration.
29. **TRUE** False Mutualism can be thought of as two species that take advantage of each other ecologically over the long term.
30. **TRUE** False The brightly colored pattern of poison dart frogs is an example of aposomatic coloration.
31. True **FALSE** Most organisms grow under idealized conditions because they are adapted to their environment.
32. **TRUE** False Population is the lowest hierarchical level for the biosphere category.
33. True **FALSE** Density-independent forces are most likely to affect population size when N is greater than K .

34. (10 points) You performed a t -test to compare a variable between two populations. The sample size from *each* population was 5 individuals. You calculated $t = 2.27$. Using $\alpha = 0.05$, do you accept or reject the null hypothesis? Explain how you used the table at right to decide, including specific numbers. **Circle the critical value of t .**

Table 1. Critical values of t .

df	α			
	0.10	0.05	0.02	0.01
1	6.31	12.71	31.82	63.66
2	2.92	4.30	6.96	9.92
3	2.35	3.18	4.54	5.84
4	2.13	2.78	3.75	4.60
5	2.02	2.57	3.36	4.03
6	1.94	2.45	3.14	3.71
7	1.89	2.36	3.00	3.50
8	1.86	2.31	2.90	3.36
9	1.83	2.26	2.82	3.25
10	1.81	2.23	2.76	3.17
11	1.80	2.20	2.72	3.11
12	1.78	2.18	2.68	3.06

Accept null hypothesis. Degrees of freedom is 8, so critical $t = 2.31$. Calculated $t = 2.27$, which is lower than the table value. If the calculated value is lower than the tabled value, then you accept (or fail to reject) the null hypothesis.

35. (10 points) Calculate the mean ($\bar{X}$) and standard deviation (s) from the five numbers listed at right. **Show your work for all steps.** If you do not show your work, you will not receive credit for this question, nor will you get any pudding. Round to one digit after the decimal for your final answers.

2	4
3	5
10	

$$\bar{X} = 4.8$$

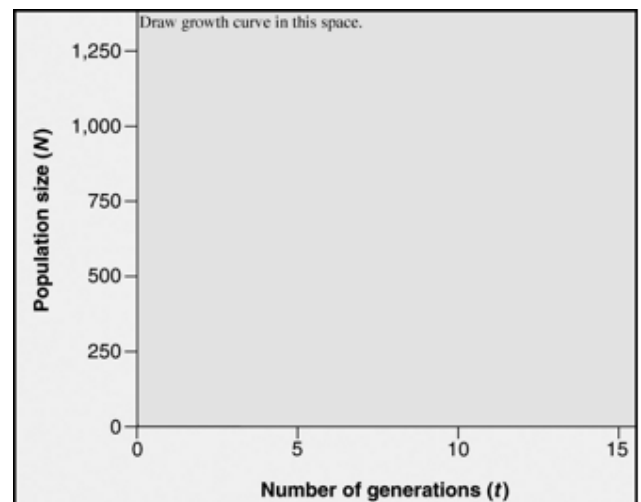
$$s = 2.8$$

$$s = \sqrt{\frac{\sum (x - \bar{X})^2}{n - 1}}$$

Give them 2 points for each correct answer, even if they do not show their work.

36. (3 points) In the space provided (right), draw all parts of the growth curve that represents the *model* of population growth when resources are *limited*. The population size that can be supported is 1000 individuals.
37. (3 points) Write the formula that represents this type of population growth.

$$\frac{\Delta N}{\Delta t} = rN \left(\frac{K - N}{K} \right)$$



38. (10 points) Consider the following metapopulation. Identify *one* subpopulation that is *most likely* to have the highest extinction rate and lowest recolonization rate. Justify your choice by explaining how patch size, distance and habitat quality interact to affect extinction and recolonization in your chosen patch. There may be more than one correct choice but explain for *only* one subpopulation.

